

System_Wasfi: Pulsed Plasma Reactor Specification

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1 System Overview

System_Wasfi is a self-sustaining, closed-loop pulsed plasma reactor. It utilizes Xenon gas as the inert plasma medium, confined within a Tungsten shell, and cooled via Tungsten Carbide internal helical piping.

2 Core Operational Parameters

The system operates on a resonant charging cycle to maintain pulse stability:

- **Fuel Mass:** 1 nanogram of Xenon
- **Pulse Energy:** $E_{pulse} = 10$ J
- **Material Integrity:** Tungsten Carbide (WC) piping for electromagnetic transparency.

3 Energy Conversion Model

The system achieves self-sustenance through a resonant circuit where the pulse frequency (f_{pulse}) is synchronized with the turbine frequency (f_{grid}):

$$f_{pulse} = \frac{f_{grid}}{N}$$

The discharge mechanism is facilitated by a fast-switching device to ensure rapid compression of the plasma within the 15 cm^3 chamber. The efficiency of heat extraction is governed by the turbulent flow within the internal cooling loops:

$$Q = \dot{m}c_p\Delta T$$

4 Conclusion

System_Wasfi represents a significant advancement in stable, pulsed energy production, shifting the focus from consumable fuel-based reactions to a robust, long-term plasma-cycle architecture.